

Daman Singh Batra

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EDUCATION

University of Alberta

Bachelor of Science with Specialization in Computing Science

- Dean's Honor Roll

Edmonton, Alberta

Expected: Dec 2027

CERTIFICATIONS

AWS Certified Solutions Architect – Associate | Amazon Web Services (AWS)

Issued: Apr 2025

EXPERIENCE

Research Assistant

University of Alberta

Sept 2026 – Present

Edmonton, AB

- Developing drift-robust anomaly detection for enterprise security logs using GNNs on provenance graphs, with game-theoretic adversarial modeling and MITRE ATT&CK-fine-tuned LLMs.

Software Engineer Intern

AltaML

May 2026 – Aug 2026

Toronto, Ontario

- Built organization-level multi-tenancy for a production Azure OpenAI RAG assistant serving government clients, isolating SQL, blob storage, and Azure AI Search indexes per tenant with cross-tenant IDOR protection to cut client provisioning from days to under an hour and per-client infrastructure cost by 30%.
- Shipped GraphRAG as a third production retrieval mode in a LangGraph workflow: schema-guided LLM entity/relation extraction over a 268-document policy corpus into a Neo4j graph of 22K entities, 37K relations and 6.1K chunks with chunk-level provenance.
- Ran LLM-as-judge evals in Arize (correctness, faithfulness, retrieval relevance) comparing GraphRAG vs. vector vs. agentic on a 194-question set; a systematic tuning sweep with semantic reranker raised answer-correctness 58.8% to 75.3%.
- Diagnosed p95 latency of 73.5s from 4,889 answer traces: isolated a post-RAG LLM judge discarding completed answers and re-running agentic search on 57% of queries; cut the agent's iteration cap 50 to 20 via a 6-config eval sweep, dropping agentic latency 28%.

Software Engineer Intern

TD Securities

Jan 2026 – Apr 2026

Toronto, Ontario

- Built real-time streaming infrastructure across 50+ Kafka topics: migrated workloads from Spark Structured Streaming to Spark Declarative Pipelines, eliminating manual state and partition management.
- Spearheaded migration of analytics pipelines from Apache Druid to Databricks Delta Lake, cutting segment extraction time by 70% by replacing sequential ZIP processing with distributed Spark jobs.
- Engineered parameterized, ADLS-integrated Databricks Jobs for parallel ingestion across high-volume tables; shipped 20+ multi-environment releases via XLR and configured networking across Databricks, Kafka, and AKS.

Software Engineer Intern

OMERS

May 2025 – Dec 2025

Toronto, Ontario

- Strengthened platform security by migrating 1,000+ monthly alert integrations from Basic Auth to OAuth 2.0, eliminating credential exposure across REST API integrations.
- Automated 20+ enterprise workflows on ServiceNow (Business Rules, UI Policies, JavaScript) and built 5+ dynamic catalog forms with REST integrations and adaptive logic, improving request fulfillment time by 30%.

PROJECTS

Tearsheet (In Progress) | Python, LangGraph, LangSmith, RAG, Claude API, pgvector, FastAPI

- Building a multi-agent due-diligence system on LangGraph: a stateful, cyclic graph orchestrating retriever, analyst, verifier, and synthesis agents with conditional routing and automated re-retrieval when claims fail validation; tracing and evaluating agent runs with LangSmith.
- Implementing RAG over SEC filings, earnings data, and news with pgvector semantic search, grounding every generated claim against source documents for citation-level traceability.

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL

AI/ML: LangGraph, LangChain, RAG (hybrid, agentic, GraphRAG), Claude API, Azure OpenAI, pgvector, Neo4j, Arize, LangSmith, OpenTelemetry, pandas, NumPy, Scikit-Learn

Real-Time & Streaming: Kafka, Spark Structured Streaming, Spark Declarative Pipelines, Databricks, Delta Lake

Frameworks: FastAPI, SQLAlchemy, React, Next.js, Node.js, Django

Data & Cloud: Azure, AWS, Spark, PostgreSQL, MSSQL, MongoDB, Docker, Kubernetes, Git, Linux